Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	16 Feb 2026
Team ID	SWTID-2026-8203
Project Name	MicroGoal
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	**User Authentication & Account Management**	User Registration
		User Login
		Password Encryption (Hashing)
		JWT Token Generation
FR-2	**Goal Management (CRUD Operations)**	Create Goal
		View All Goals (User-Specific)
		Update Goal
		Delete Goal
FR-3	**Authorization & Access Control**	Protect Routes using Auth Middleware

		Restrict Users from Accessing Others' Data
FR-4	**Error Handling & Response Management**	Centralized Error Middleware
		Proper HTTP Status Codes (200, 201, 400, 401, 404, 500)
FR-5	**API Testing & Validation**	Test Endpoints using Thunder Client
		Validate Request Body Data

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	API endpoints should be simple, clear, and easy to understand for developers.
NFR-2	**Security**	All API requests must be secured using JWT authentication. Passwords must be hashed before storing.
NFR-3	**Reliability**	The system should handle errors properly and avoid crashes during invalid requests.
NFR-4	**Performance**	API responses should be returned within 2 seconds under normal load conditions.

NFR-5	**Availability**	The backend server should be accessible 24/7 with minimal downtime.
NFR-6	**Scalability**	The system should support increasing users and goals without performance degradation. The architecture should allow future feature expansion.